

Math and Computational Biology I

Sep 28 2026

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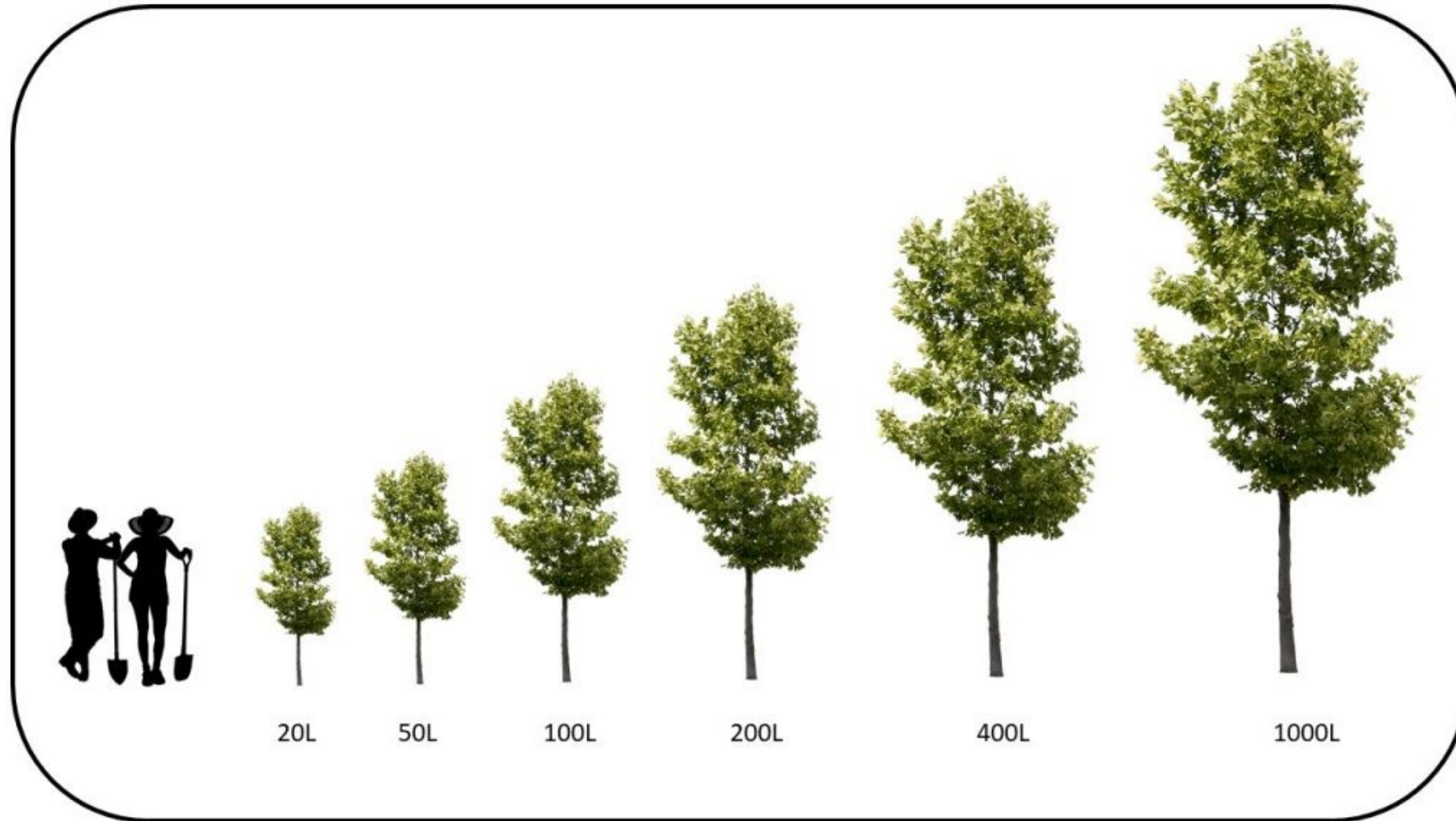
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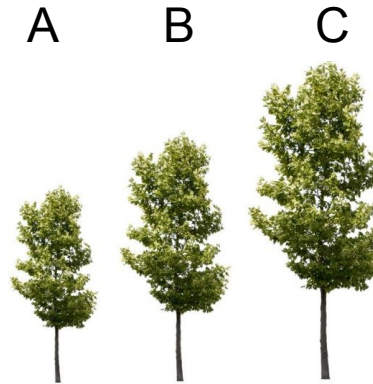
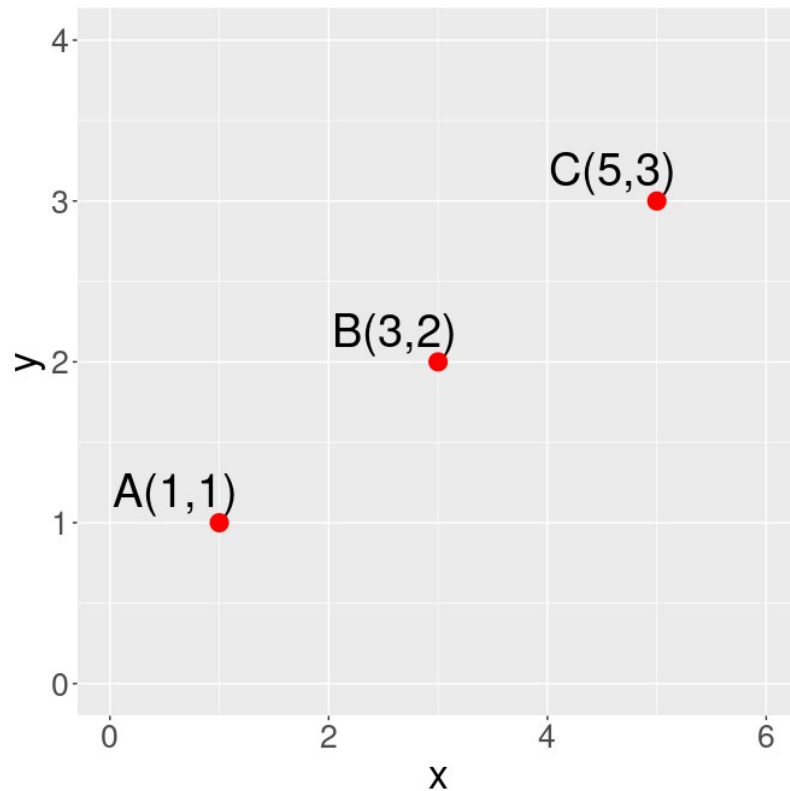
Content

- Derive Simple Linear Regression
 - Linear Algebra
 - Calculus
 - Statistics
- Algorithms to find the solution for Simple Linear regression
 - Gradient descent
 - Pseudo Inverse
- Multiple Linear Regression
- Related linear regression

Prediction weight of tree using age: Simple Linear Regression



Fitting a line going through three points (1): Linear Algebra



	x (Age)	y (Weight)
A	1	1
B	3	2
C	5	3

How to find the line going through A and B?

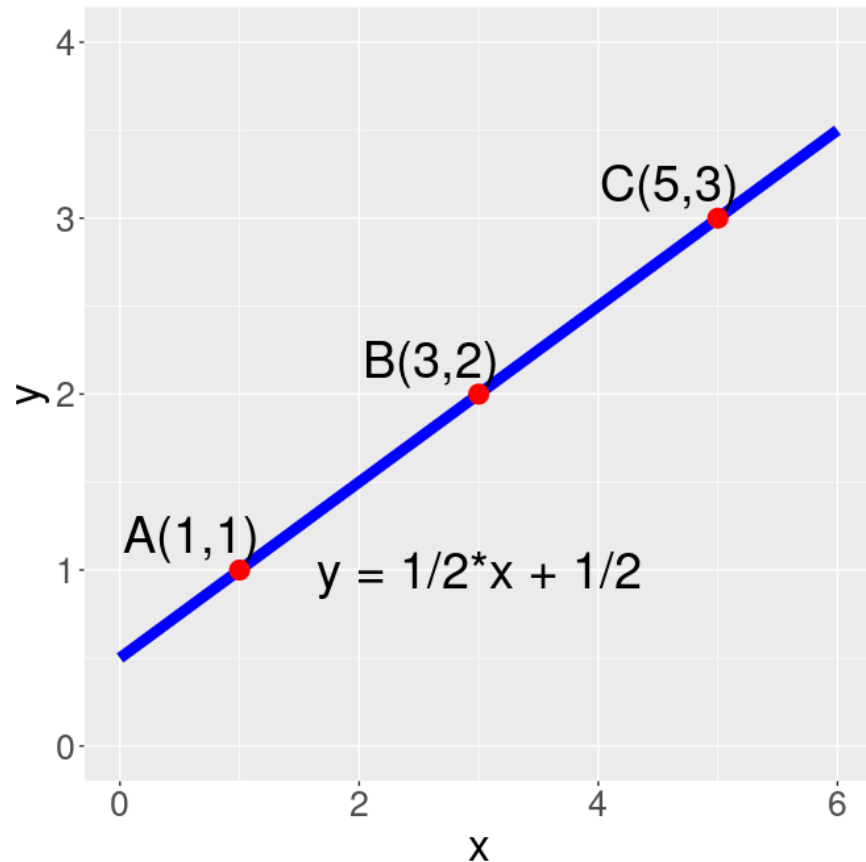
$$y = f(x) = m \cdot x + c$$

```
library(ggplot2)
```

```
## plot 3 points
```

```
p <- ggplot(data=data.frame(x=0, y=0), mapping = aes(x=x, y=y)) +  
  geom_text(x=1-0.4, y=1+0.2, label="A(1,1)", size=10) +  
  geom_text(x=3-0.4, y=2+0.2, label="B(3,2)", size=10) +  
  geom_text(x=5-0.4, y=3+0.2, label="C(5,3)", size=10) +  
  geom_point(x=1, y=1, color="red", size=5) +  
  geom_point(x=3, y=2, color="red", size=5) +  
  geom_point(x=5, y=3, color="red", size=5) +  
  xlim(0,6) + ylim(0, 4) +  
  theme(text = element_text(size = 25))
```

Fitting a line going through three points (2)



	x (Age)	y (Weight)
A	1	1
B	3	2
C	5	3

Find the line going through A and B: $y = f(x) = m * x + c$

For A: $1 = m * 1 + c$ (1)

For B: $2 = m * 3 + c$ (2)

For C: $3 = m * 5 + c$ (3)

Solving system of 2 equations (1) and (2):

$m = 1/2$

$c = 1/2$

replace values of m, c to (3): $3 = 1/2 * 5 + 1/2$ (TRUE)

$\Rightarrow y = 1/2 * x + 1/2$ with $m = 1/2$ and $c = 1/2$

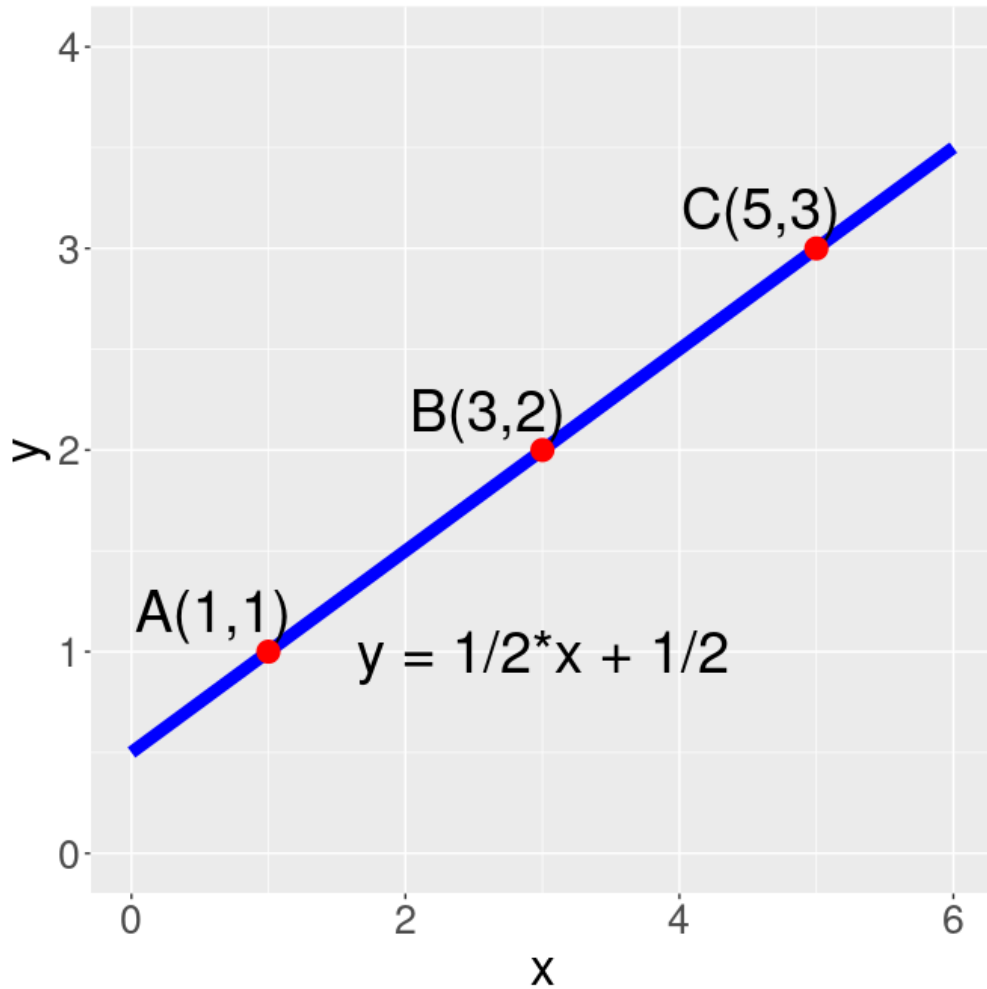
```
library(ggplot2)
```

```
## plot line going through 3 points
```

```
l2 <- function(x){ 1/2*x + 1/2 }
```

```
p <- ggplot(data = data.frame(x = 0), mapping = aes(x = x)) +  
  stat_function(fun = l2, color = "blue", size = 3) + xlim(0, 6) + ylim(0, 4) +  
  geom_text(x = 1 - 0.4, y = 1 + 0.2, label = "A(1,1)", size = 10) +  
  geom_text(x = 3 - 0.4, y = 2 + 0.2, label = "B(3,2)", size = 10) +  
  geom_text(x = 5 - 0.4, y = 3 + 0.2, label = "C(5,3)", size = 10) +  
  geom_point(x = 1, y = 1, color = "red", size = 5) +  
  geom_point(x = 3, y = 2, color = "red", size = 5) +  
  geom_point(x = 5, y = 3, color = "red", size = 5) +  
  geom_text(x = 3, y = 1, label = "y = 1/2 * x + 1/2", size = 10) +  
  theme(text = element_text(size = 25))
```

Fitting a line going through three points (3)



	x (Age)	y (Weight)
A	1	1
B	3	2
C	5	3

Find the line going through A and B:

$$y = m * x + c$$

For A: $1 = m * 1 + c$

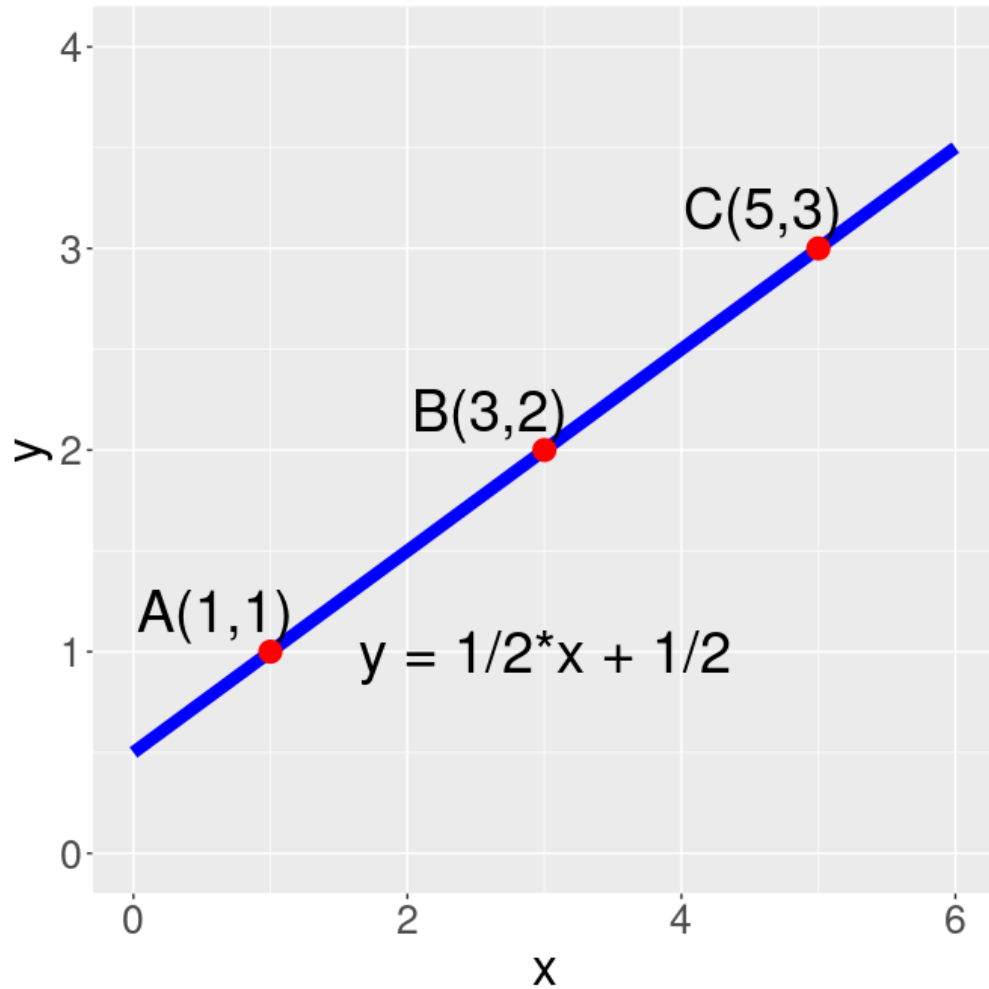
For B: $2 = m * 3 + c$

For C: $3 = m * 5 + c$

Rewrite system of 3 equations into matrix format

$$Ax = b$$

Fitting a line going through three points (4)



	x (Age)	y (Weight)
A	1	1
B	3	2
C	5	3

Find the line going through A and B:

$$y = m * x + c$$

For A: $1 = m * 1 + c$

For B: $2 = m * 3 + c$

For C: $3 = m * 5 + c$

Rewrite system of 3 equations into matrix format

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$Ax = b$

Fitting a line going through two points (5)

	x (Age)	y (Weight)
A	1	1
B	3	2

$$X1 + X2 = 1$$

$$3X1 + X2 = 2$$

I) Row Space

1) Substitution Method

2) Gaussian Elimination

3) Plot

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

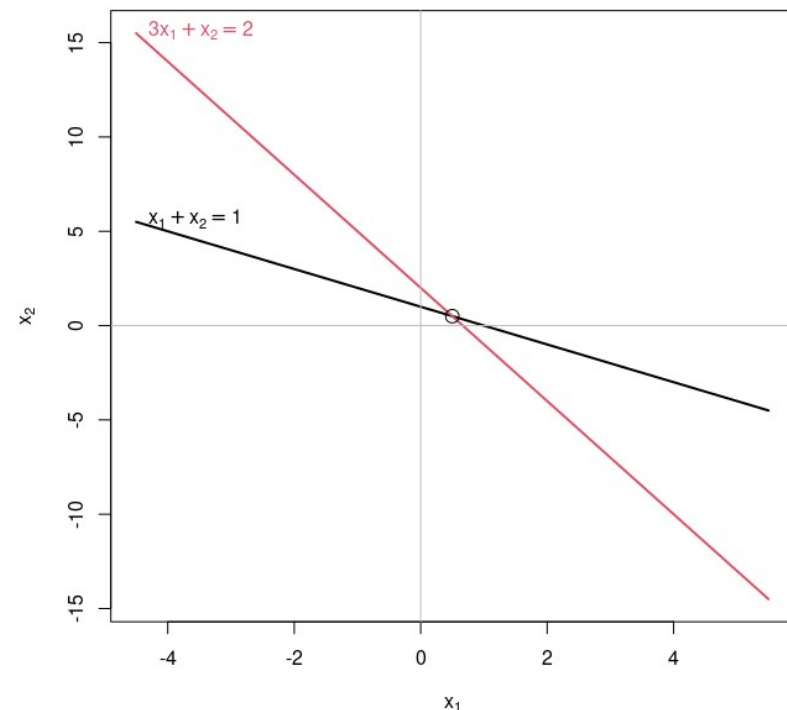
$Ax = b$

$$\begin{aligned} X1 + X2 &= 1 \\ 3X1 + X2 &= 2 \end{aligned}$$

```
> A <- matrix(c(1,3,1,1), nrow=2, ncol=2)
> A
      [,1] [,2]
[1,]    1    1
[2,]    3    1
> b <- c(1,2)
> b
[1] 1 2
> solve(A,b)
[1] 0.5 0.5
> plotEqn(A,b)
      x[1] + x[2] = 1
    3*x[1] + x[2] = 2
```

$$m=0.5$$

$$c=0.5$$

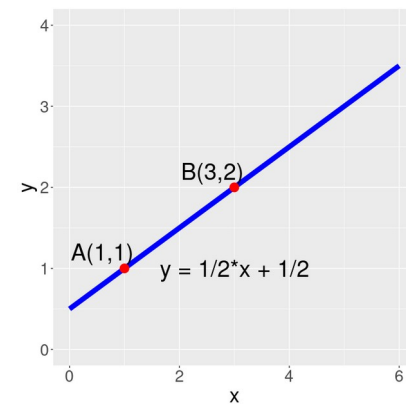
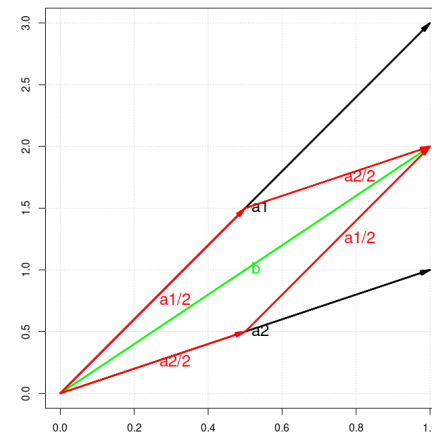
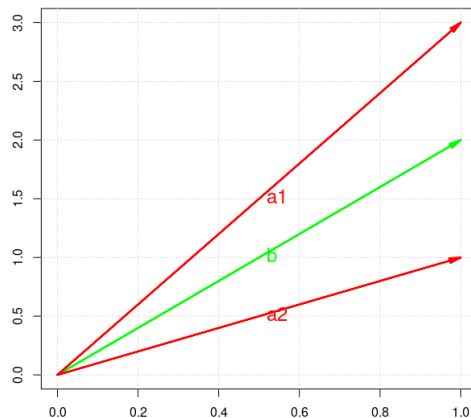


II) Column Space

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} m + \begin{bmatrix} 1 \\ 1 \end{bmatrix} c = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$m=0.5$$

$$c=0.5$$



Fitting a line going through two points (5): Row Space

	x (Age)	y (Weight)
A	1	1
B	3	2

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$Ax = b$

$$\begin{aligned} X_1 + X_2 &= 1 \quad (1) \\ 3X_1 + X_2 &= 2 \quad (2) \end{aligned}$$

1) Substitution Method

$$(1) \rightarrow X_1 = 1 - X_2$$

$$\begin{aligned} \text{Substitute to (2):} \\ 3(1 - X_2) + X_2 &= 2 \\ 3 - 3X_2 + X_2 &= 2 \\ 3 - 2X_2 &= 2 \\ X_2 &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{Substitute to (1):} \\ X_1 = 1 - X_2 &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} X_1 + X_2 &= 1 \quad (1) \\ 3X_1 + X_2 &= 2 \quad (2) \end{aligned}$$

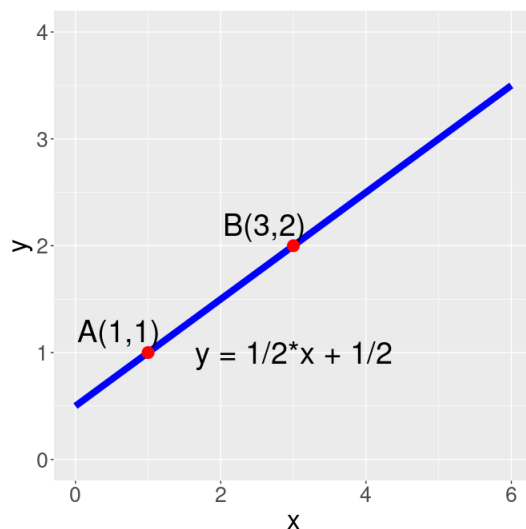
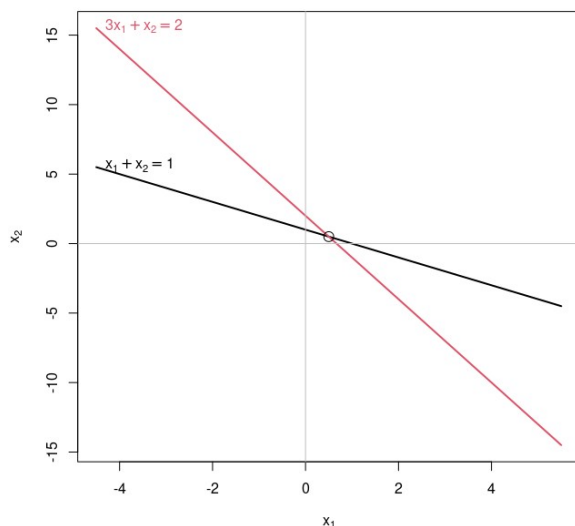
2) Gaussian Elimination

$$\begin{aligned} \text{1) Multiply both side of (1) with -1} \\ -X_1 - X_2 &= -1 \quad (1') \\ \text{2) (1') + (2):} \\ -X_1 - X_2 + 3X_1 + X_2 &= -1 + 2 \\ 2X_1 &= 1 \\ X_1 &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{Substitute to (1):} \\ X_2 = 1 - X_1 &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} X_1 + X_2 &= 1 \quad (1) \\ 3X_1 + X_2 &= 2 \quad (2) \end{aligned}$$

3) Plot
 $X_1 = X_2 = \frac{1}{2}$



Fitting a line going through two points (5): Column Space

	x (Age)	y (Weight)
A	1	1
B	3	2

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

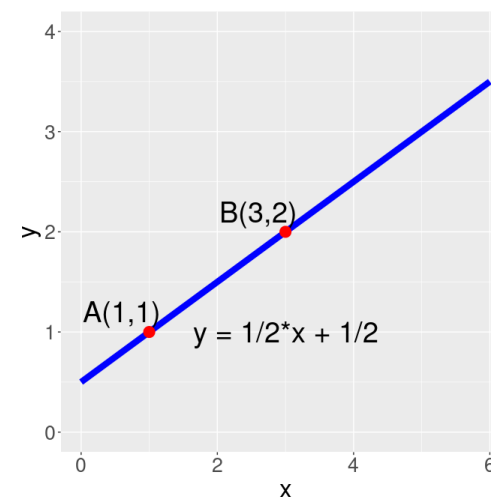
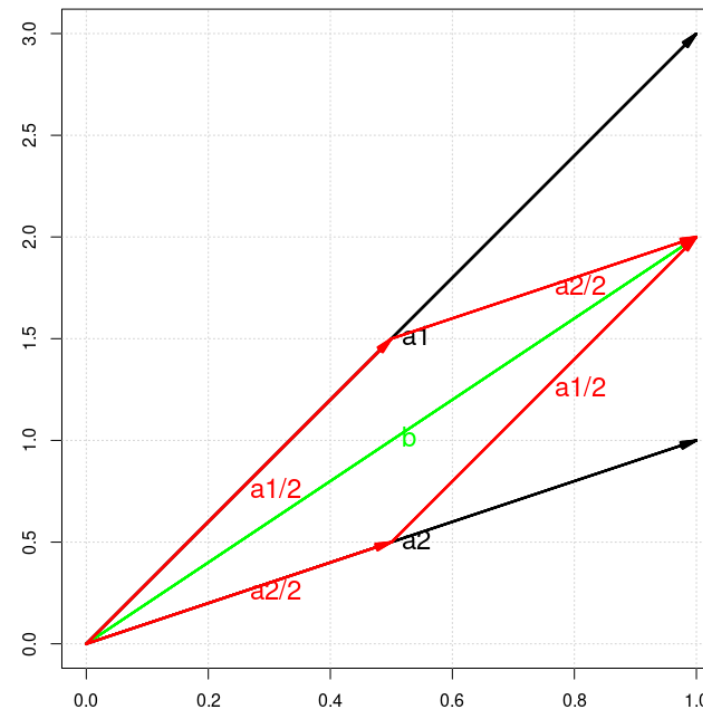
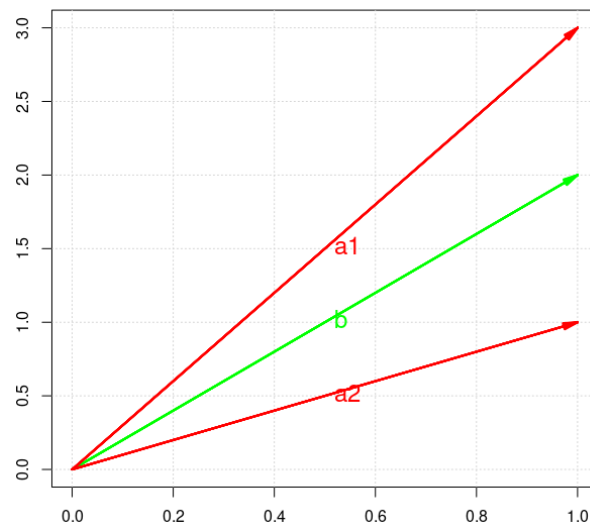
$Ax = b$

$$\begin{aligned} X_1 + X_2 &= 1 \\ 3X_1 + X_2 &= 2 \end{aligned}$$

II) Column Space

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} m + \begin{bmatrix} 1 \\ 1 \end{bmatrix} c = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

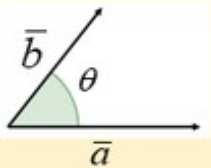
$$\begin{aligned} m &= 0.5 \\ c &= 0.5 \end{aligned}$$



Multiplying vectors: The dot (scalar/inner) product

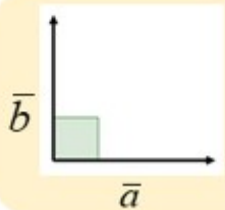
If $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$
then the dot product is

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$



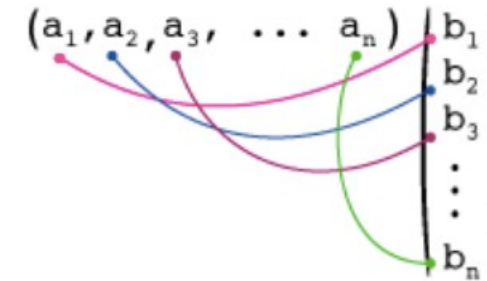
If θ is the angle between $\vec{a}$ and $\vec{b}$ then

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$



$\vec{a} \cdot \vec{b}$ are orthogonal (perpendicular)

if and only if $\vec{a} \cdot \vec{b} = 0$



An example

Here's an example of the dot product of two 3-D vectors. Notice that it wouldn't make any sense to multiply an n -dimensional vector by an m -dimensional vector. The number of elements of the vectors must match in order for us to find a dot product.

$$\mathbf{a} = (1, 4, -2)$$

$$\mathbf{b} = (-2, 1, 7)$$

$$\mathbf{a} \cdot \mathbf{b} = 1 \cdot (-2) + 4 \cdot 1 + (-2) \cdot 7$$

$$= -2 + 4 - 14 = -12$$

A 3×3 matrix multiplied by a 3×1 vector

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 + 1 \cdot 1 + 1 \cdot 2 \\ 2 \cdot 3 + 1 \cdot 1 + 3 \cdot 2 \\ 1 \cdot 3 + 4 \cdot 1 + 2 \cdot 2 \end{pmatrix} \quad \text{First row,}$$

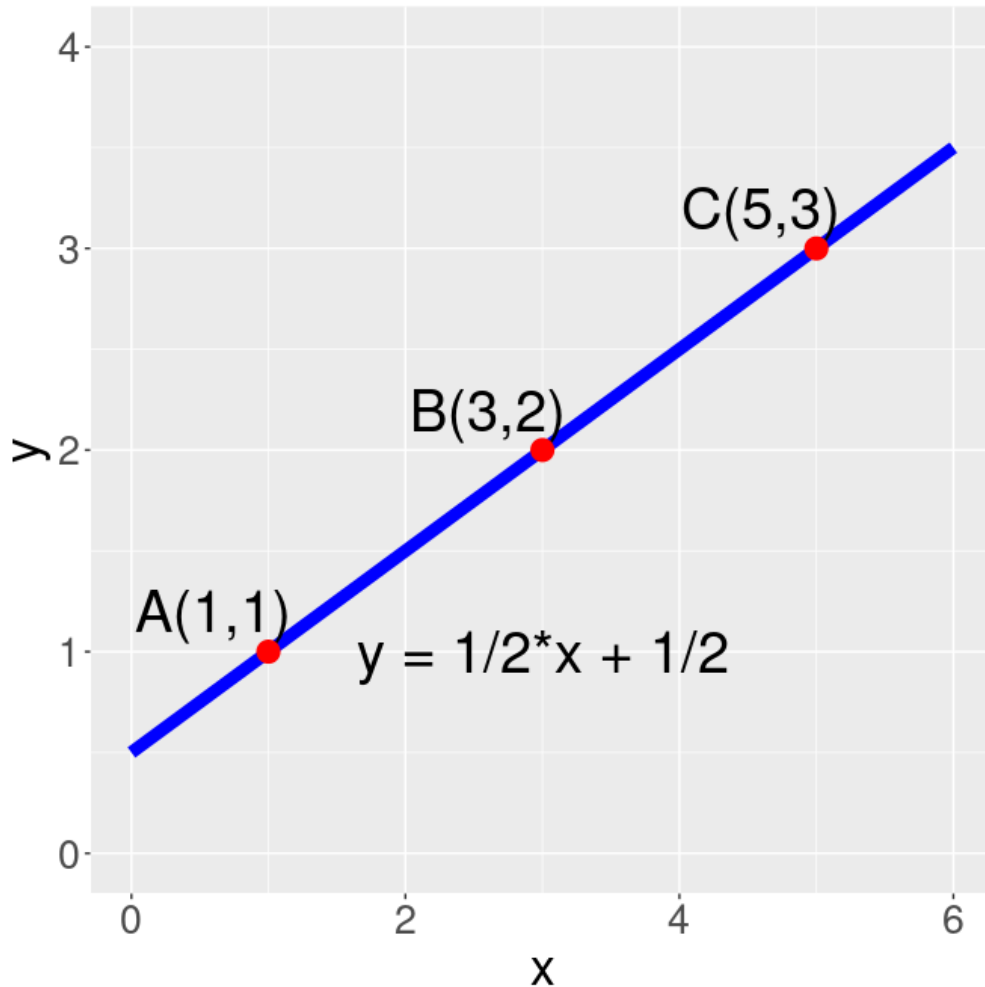
$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 + 1 \cdot 1 + 1 \cdot 2 \\ 2 \cdot 3 + 1 \cdot 1 + 3 \cdot 2 \\ 1 \cdot 3 + 4 \cdot 1 + 2 \cdot 2 \end{pmatrix} \quad \text{next row,}$$

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 + 1 \cdot 1 + 1 \cdot 2 \\ 2 \cdot 3 + 1 \cdot 1 + 3 \cdot 2 \\ 1 \cdot 3 + 4 \cdot 1 + 2 \cdot 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 13 \\ 11 \end{pmatrix} \quad \begin{array}{l} \text{last row,} \\ \text{then do the addition.} \end{array}$$

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$Ax = y$

Fitting a line going through multiple points with one feature



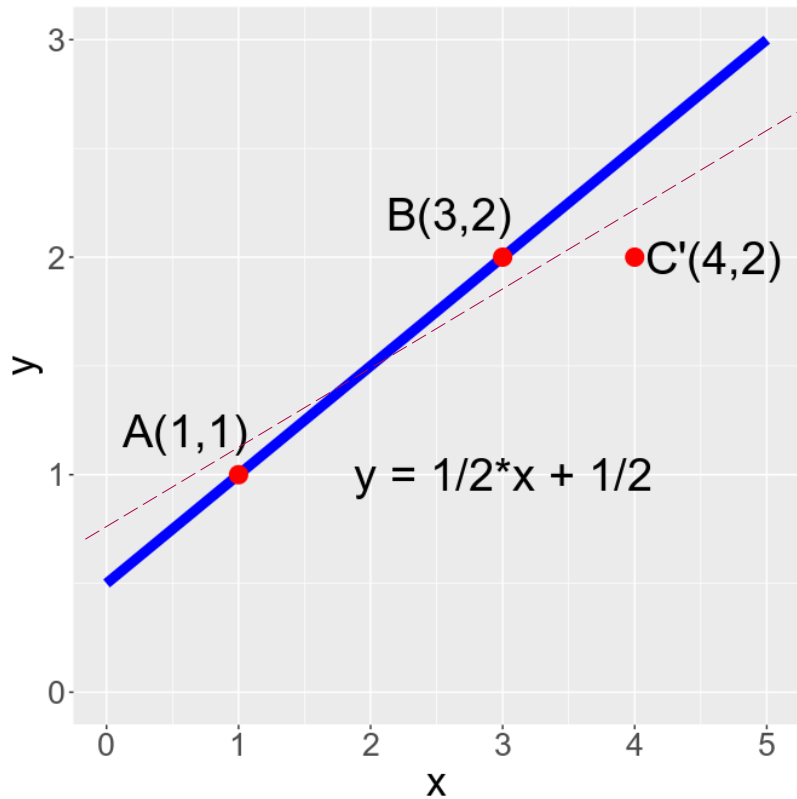
Plant	x (Age, month) independent variable/predictor/ feature/attribute	y (Weight, gram) dependent/Predicted variable/class
A	1	1
B	3	2
C	5	3
...	...	...

Matrix format:

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 5 & 1 \\ \dots & \dots \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ \dots \end{bmatrix}$$

$Ax = b$

Cannot Fitting a line going through three points



	x	y
A	1	1
B	3	2
C'	4	2

There is not a line going through A, B and C':

$$y = m * x + c$$

For A: $1 = m * 1 + c$

For B: $2 = m * 3 + c$

For C': $2 = m * 4 + c$

=> Why ?

Rewrite system of 3 equations into matrix format:

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$Ax = b$

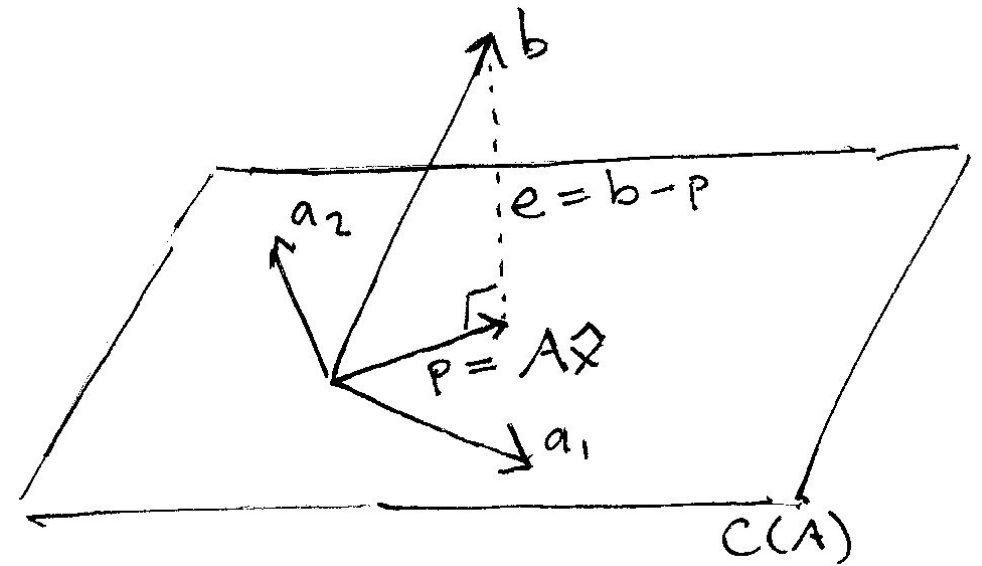
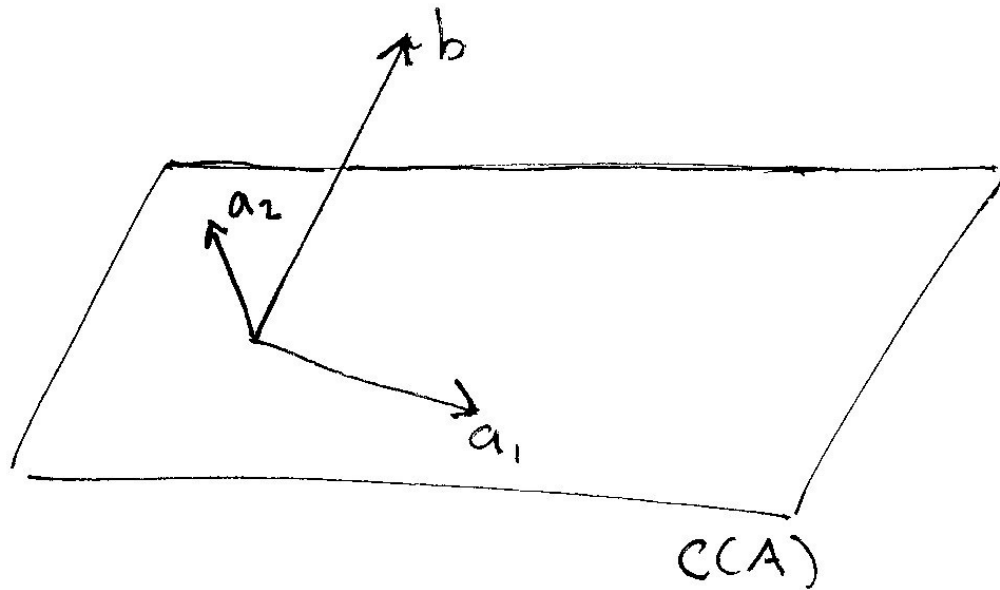
Cannot Fitting a line going through three points

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} m + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} c = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

a1 a2 b

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$Ax = b$



If we have a vector $\mathbf{b}$ and a line determined by a vector $\mathbf{a}$, how do we find the point on the line that is closest to $\mathbf{b}$?

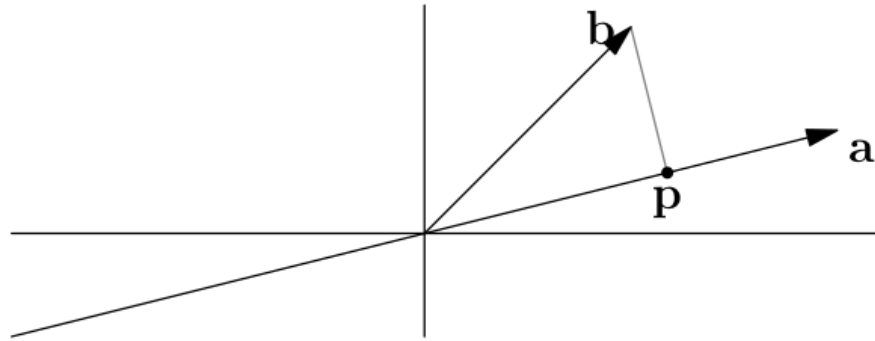


Figure 1: The point closest to $\mathbf{b}$ on the line determined by $\mathbf{a}$.

We can see from Figure 1 that this closest point $\mathbf{p}$ is at the intersection formed by a line through $\mathbf{b}$ that is orthogonal to $\mathbf{a}$. If we think of $\mathbf{p}$ as an approximation of $\mathbf{b}$, then the length of $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is the error in that approximation.

We could try to find $\mathbf{p}$ using trigonometry or calculus, but it's easier to use linear algebra. Since $\mathbf{p}$ lies on the line through $\mathbf{a}$, we know $\mathbf{p} = x\mathbf{a}$ for some number x . We also know that $\mathbf{a}$ is perpendicular to $\mathbf{e} = \mathbf{b} - x\mathbf{a}$:

$$\begin{aligned} \mathbf{a}^T(\mathbf{b} - x\mathbf{a}) &= 0 \\ x\mathbf{a}^T\mathbf{a} &= \mathbf{a}^T\mathbf{b} \\ x &= \frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}, \end{aligned}$$

and $\mathbf{p} = \mathbf{a}x = \mathbf{a} \frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}$. Doubling $\mathbf{b}$ doubles $\mathbf{p}$. Doubling $\mathbf{a}$ does not affect $\mathbf{p}$.

We'd like to write this projection in terms of a *projection matrix* P : $\mathbf{p} = P\mathbf{b}$.

$$\mathbf{p} = x\mathbf{a} = \frac{\mathbf{a}\mathbf{a}^T\mathbf{a}}{\mathbf{a}^T\mathbf{a}},$$

so the matrix is:

$$P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T\mathbf{a}}.$$

Note that $\mathbf{a}\mathbf{a}^T$ is a three by three matrix, not a number; matrix multiplication is not commutative.

$$\mathbf{a}^T \cdot \mathbf{e} = 0$$

$$\mathbf{a}^T \cdot (\mathbf{b} - x\mathbf{a}) = 0$$

$$1 \times 3 \quad 3 \times 1$$

$$3 \times 1 \quad 1 \times 3$$

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \mathbf{m} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \mathbf{c} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

a1 a2 b

$$Ax=b$$

$$A^T Ax=A^T b$$

Projection in higher dimensions

In $\mathbb{R}^3$, how do we project a vector $\mathbf{b}$ onto the closest point $\mathbf{p}$ in a plane?

If $\mathbf{a}_1$ and $\mathbf{a}_2$ form a basis for the plane, then that plane is the column space of the matrix $A = [\mathbf{a}_1 \quad \mathbf{a}_2]$.

We know that $\mathbf{p} = \hat{x}_1 \mathbf{a}_1 + \hat{x}_2 \mathbf{a}_2 = A\hat{\mathbf{x}}$. We want to find $\hat{\mathbf{x}}$. There are many ways to show that $\mathbf{e} = \mathbf{b} - \mathbf{p} = \mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to the plane we're projecting onto, after which we can use the fact that $\mathbf{e}$ is perpendicular to $\mathbf{a}_1$ and $\mathbf{a}_2$:

$$\mathbf{a}_1^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \quad \text{and} \quad \mathbf{a}_2^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0.$$

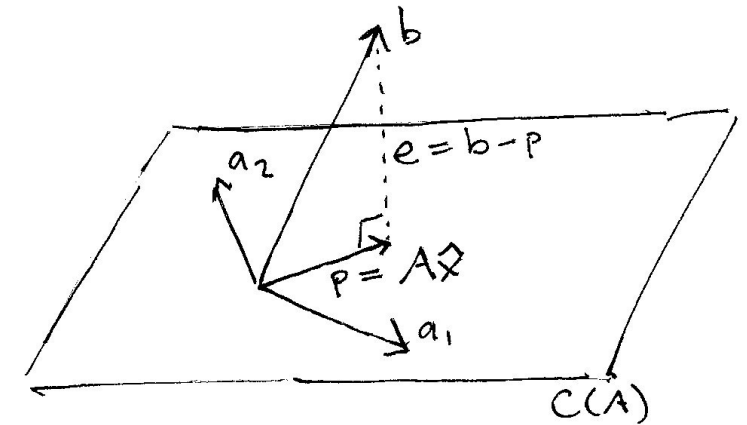
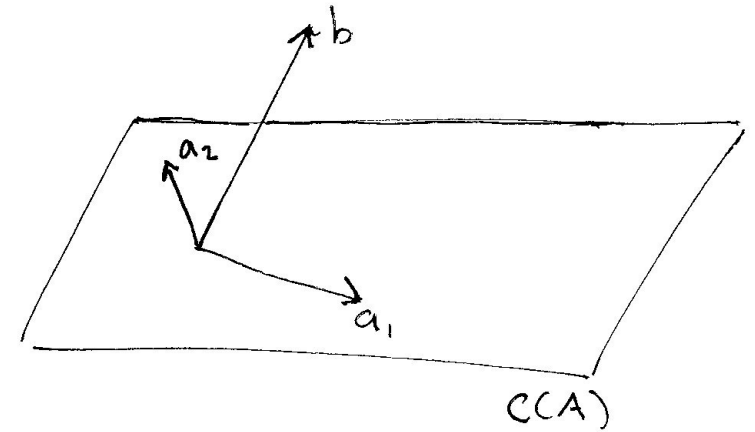
In matrix form, $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$. When we were projecting onto a line, A only had one column and so this equation looked like: $a^T(\mathbf{b} - xa) = 0$.

Note that $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ is in the nullspace of A^T and so is in the left nullspace of A . We know that everything in the left nullspace of A is perpendicular to the column space of A , so this is another confirmation that our calculations are correct.

We can rewrite the equation $A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}$ as:

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

When projecting onto a line, $A^T A$ was just a number; now it is a square matrix. So instead of dividing by $\mathbf{a}^T \mathbf{a}$ we now have to multiply by $(A^T A)^{-1}$



Exercise: Cannot Fitting a line going through three points

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} m + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} c = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

a1 a2 b

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$Ax = b$

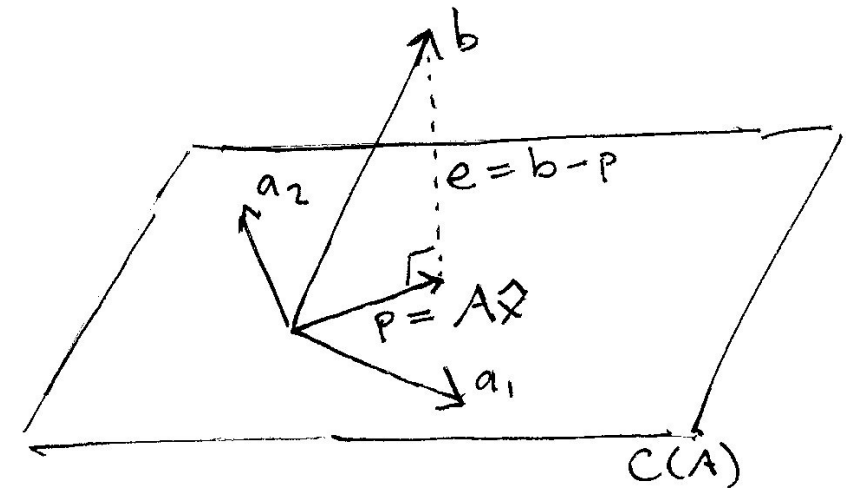
- Compute $A^T A$, $A^T b$ to find x in the normal equation

$$A^T A \hat{x} = A^T b.$$

```
> A <- matrix(c(1,3,4,1,1,1),3,2)
> A
     [,1] [,2]
[1,]    1    1
[2,]    3    1
[3,]    4    1
> m
[1] 0.3571429
> t(A)%*%A
     [,1] [,2]
[1,]   26    8
[2,]    8    3
> t(A)%*%b
     [,1]
[1,]   15
[2,]    5
```

$$A^T A = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 26 & 8 \\ 8 & 3 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 3 & 4 \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$



Exercise: Cannot Fitting a line going through three points

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} m + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} c = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

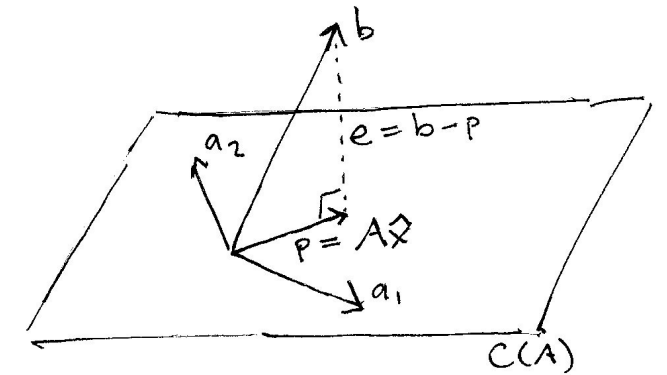
a1 a2 b

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$Ax = b$

- Compute $A^T A$, $A^T b$ to find $\hat{x}$ in the normal equation

$$A^T A \hat{x} = A^T b.$$

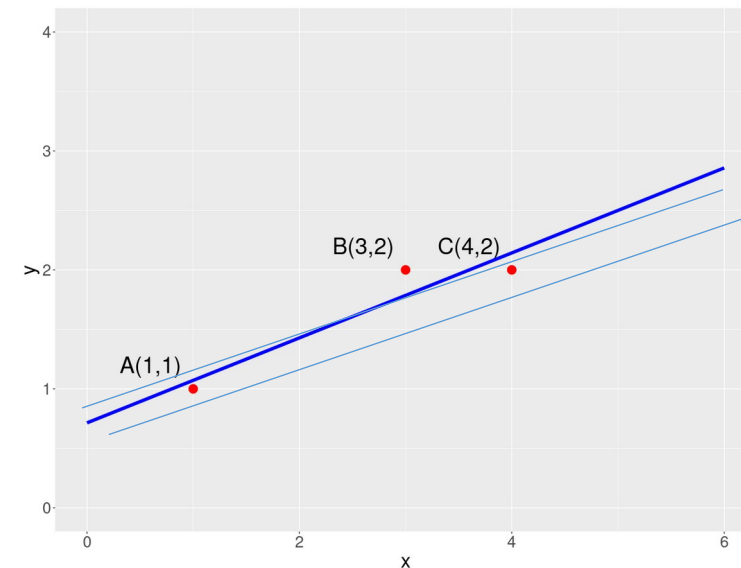


$$A^T A \hat{x} = A^T b$$

$$\Rightarrow \begin{bmatrix} 26 & 8 \\ 8 & 3 \end{bmatrix} \hat{x} = \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 26 & 8 \\ 8 & 3 \end{bmatrix} \begin{bmatrix} \hat{m} \\ \hat{c} \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \hat{m} = 0.357 \\ \hat{c} = 0.714 \end{cases}$$



Exercise: using R to check the result

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} m + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} c = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

a1 a2 b

- `A <-`
- `b <-`
- `lm(?~?)`

```
> A <- matrix(c(1,3,4,1,1,1),3,2)
> b <- c(1,2,2)
> lm(b~A)
```

Call:
lm(formula = b ~ A)

Coefficients:
(Intercept) A1 A2
0.7143 0.3571 NA

$$\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

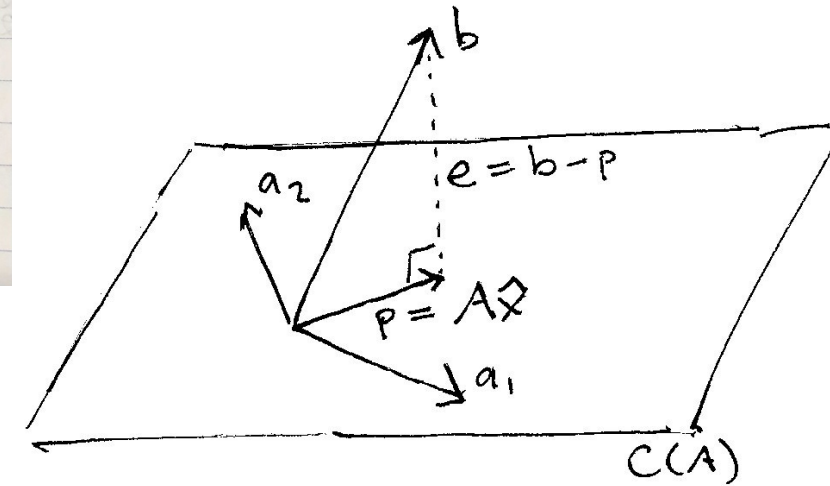
$Ax = b$

$$A^T A \bar{x} = A^T b$$

$$\Rightarrow \begin{bmatrix} 26 & 8 \\ 8 & 3 \end{bmatrix} \bar{x} = \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 26 & 8 \\ 8 & 3 \end{bmatrix} \begin{bmatrix} \hat{m} \\ \hat{c} \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \hat{m} = 0.357 \\ \hat{c} = 0.714 \end{cases}$$



Cannot Fitting a line going through three points: using calculus

$$\begin{matrix} \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} m + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} c = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \\ a1 \qquad \qquad a2 \qquad \qquad b \end{matrix}$$

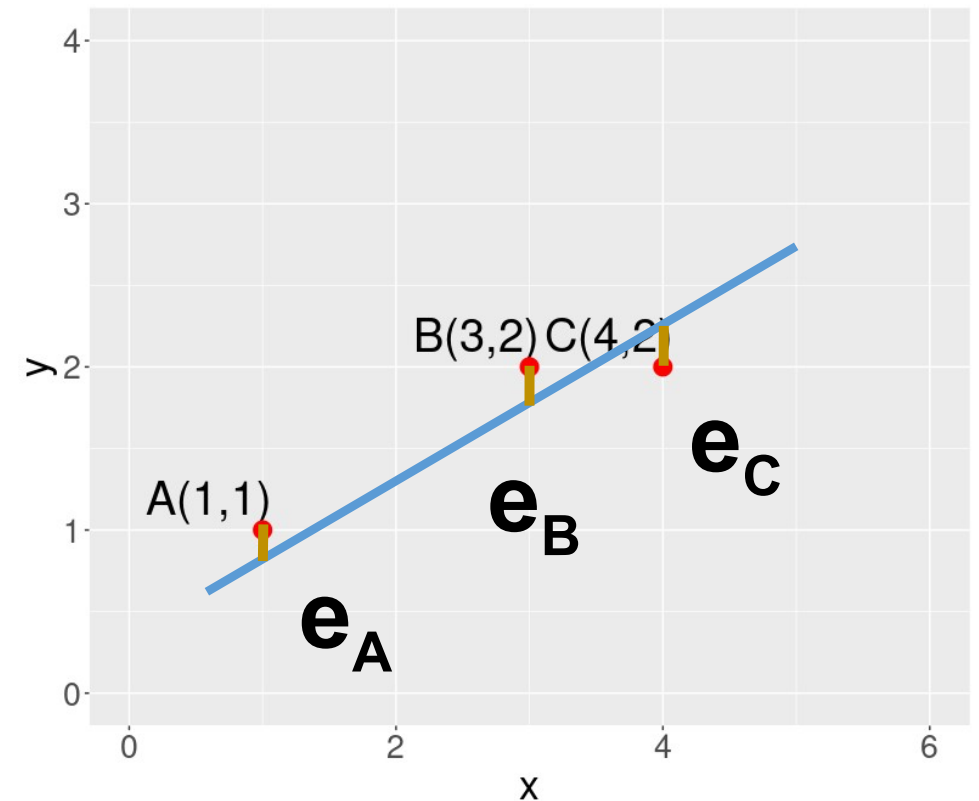
$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$Ax = b$

- **There is not** a line going through A, B and C':

$$y = m * x + c$$

- For A: $1 = m * 1 + c$
- For B: $2 = m * 3 + c$
- For C': $2 = m * 4 + c$
- Minimize sum of square of e_A, e_B, e_C



Exercise: Minimize sum of square of e_A , e_B , e_C

- **There is not** a line going through A, B and C':

$$y = m * x + c$$

- For A: $1 = m * 1 + c$
- For B: $2 = m * 3 + c$
- For C': $2 = m * 4 + c$

$$A^T A \vec{x} = A^T b$$

$$\Leftrightarrow \begin{bmatrix} 26 & 8 \\ 8 & 3 \end{bmatrix} \vec{x} = \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 26 & 8 \\ 8 & 3 \end{bmatrix} \begin{bmatrix} \hat{m} \\ \hat{c} \end{bmatrix} = \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\Rightarrow \begin{cases} \hat{m} = 0.357 \\ \hat{c} = 0.714 \end{cases}$$

$$\begin{bmatrix} 1 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$Ax = b$

- $e_A^2 = (y_{Aline} - y_A)^2 = (m * 1 + c - 1)^2 = (m + c - 1)^2$
- $e_B^2 = (y_{Bline} - y_B)^2 = (m * 3 + c - 2)^2 = (3m + c - 2)^2$
- $e_C^2 = (y_{Cline} - y_C)^2 = (m * 4 + c - 2)^2 = (4m + c - 2)^2$
- $e_{total}^2 = e_A^2 + e_B^2 + e_C^2$

$$= (m + c - 1)^2 + (3m + c - 2)^2 + (4m + c - 2)^2$$

$$= c^2 + m^2 - 2m + 2cm - 2c + 1 + c^2 + 9m^2 - 12m + 6cm - 4c$$

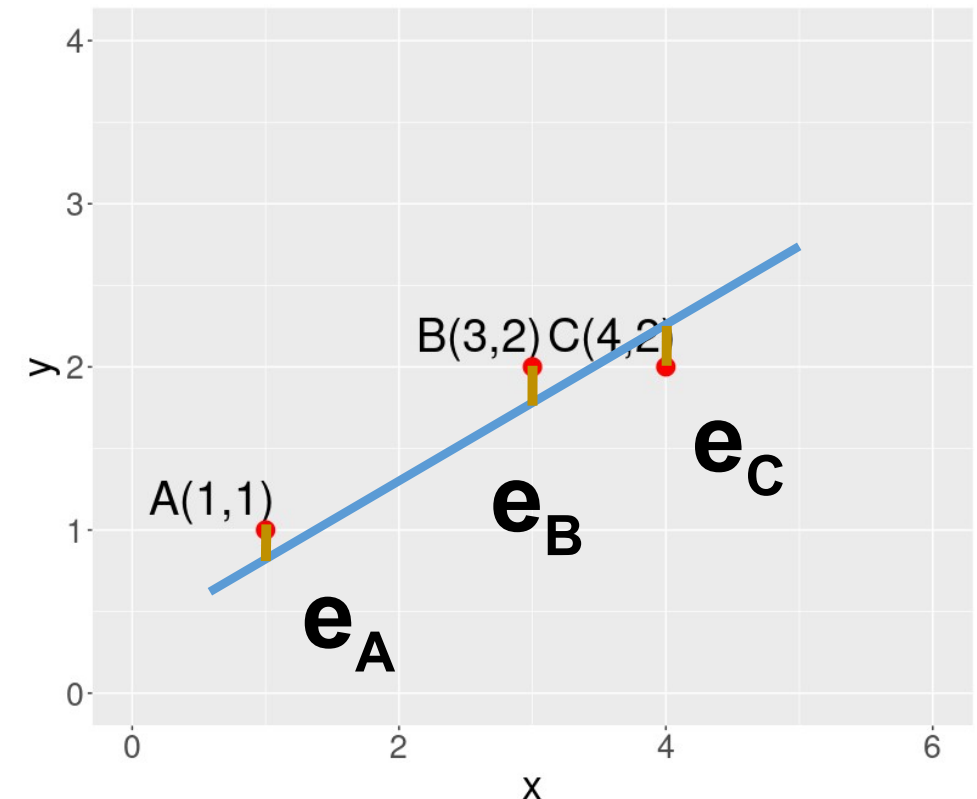
$$+ 4 + c^2 + 16m^2 - 16m + 8cm - 4c + 4$$

$$= 26m^2 + 3c^2 - 30m + 16cm - 10c + 9$$

$$de_{total}^2 / dm = 52m + 16c - 30 = 0$$

$$de_{total}^2 / dc = 16m + 6c - 10 = 0$$

$$\rightarrow m = 5/14 = 0.357 \text{ and } c = 5/7 = 0.714$$



Linear regression is equivalent to doing MLE on the Gaussian target

We start with the statistical model, which is the Gaussian-noise simple linear regression model, defined as follows:

1. The distribution of X is arbitrary (and perhaps X is even non-random).
2. If $X = x$, then $Y = \beta_0 + \beta_1 x + \epsilon$, for some constants (“coefficients”, “parameters”) β_0 and β_1 , and some random noise variable ϵ .
3. $\epsilon \sim N(0, \sigma^2)$, and is independent of X .
4. ϵ is independent across observations.

A consequence of these assumptions is that the response variable Y is independent across observations, conditional on the predictor X , i.e., Y_1 and Y_2 are independent given X_1 and X_2 (Exercise 1).

As you’ll recall, this is a special case of the simple linear regression model: the first two assumptions are the same, but we are now assuming much more about the noise variable ϵ : it’s not just mean zero with constant variance, but it has a particular distribution (Gaussian), and everything we said was uncorrelated before we now strengthen to independence¹.

Because of these stronger assumptions, the model tells us the conditional pdf of Y for each x , $p(y|X = x; \beta_0, \beta_1, \sigma^2)$. (This notation separates the random variables from the parameters.) Given any data set $(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$, we can now write down the probability density, under the model, of seeing that data:

$$\prod_{i=1}^n p(y_i|x_i; \beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_i))^2}{2\sigma^2}}$$

$$\text{observed data} \rightarrow y = b_0 + b_1 x + \epsilon$$

$$\text{predicted data} \rightarrow y' = b_0 + b_1 x$$

$$\text{error} \rightarrow \epsilon = y - y'$$

In multiplying together the probabilities like this, we are using the independence of the Y_i .

When we see the data, we do not *known* the true parameters, but any guess at them, say (b_0, b_1, s^2) , gives us a probability density:

$$\prod_{i=1}^n p(y_i|x_i; b_0, b_1, s^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi s^2}} e^{-\frac{(y_i - (b_0 + b_1 x_i))^2}{2s^2}}$$

This is the **likelihood**, a function of the parameter values. It’s just as informative, and much more convenient, to work with the **log-likelihood**,

$$L(b_0, b_1, s^2) = \log \prod_{i=1}^n p(y_i|x_i; b_0, b_1, s^2) \quad (1)$$

$$= \sum_{i=1}^n \log p(y_i|x_i; b_0, b_1, s^2) \quad (2)$$

$$= -\frac{n}{2} \log 2\pi - n \log s - \frac{1}{2s^2} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2 \quad (3)$$

Linear regression is equivalent to doing MLE on the Gaussian target

Linear regression is the basic model for regression analysis; its simplicity allows us to apply analytical methods. We often define the true regression value $\hat{y}$ following the Gaussian distribution:

$$\hat{y} \sim \mathcal{N}(W^T x, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\hat{y}-W^T x)^2}{2\sigma^2}}$$

where $W^T x$ is the predicted value from linear regression. We can perform both MLE and MAP analytically.

MLE

$$\begin{aligned} W_{MLE} &= \operatorname{argmax}_W \log \hat{y} \\ &= \operatorname{argmax}_W \log \frac{1}{\sqrt{2\pi}\sigma} + \log \left(\exp \left(-\frac{(\hat{y}-W^T x)^2}{2\sigma^2} \right) \right) \\ &= \operatorname{argmax}_W -\frac{(\hat{y}-W^T x)^2}{2\sigma^2} - \log \sigma \\ &= \operatorname{argmin}_W \frac{1}{2}(\hat{y}-W^T x)^2 \quad \text{Regard } \sigma \text{ as constant} \end{aligned}$$

We can see that if we regard the variance σ^2 as constant, then linear regression is equivalent to doing MLE on the Gaussian target.

MAP is equivalent to the linear regression with L2/ridge regularization

MAP

As we already know, MAP has an additional priori than MLE. We assume the prior distribution $P(W)$ as Gaussian distribution $\mathcal{N}(0, \sigma_0^2)$ as well:

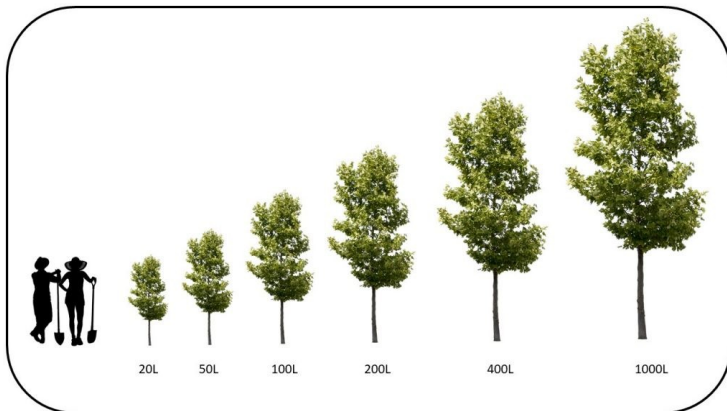
$$\begin{aligned} W_{MAP} &= \operatorname{argmax}_W W_{MLE} + \log P(W) \\ &= \operatorname{argmax}_W W_{MLE} + \log \mathcal{N}(0, \sigma_0^2) \\ &= \operatorname{argmax}_W W_{MLE} + \log \exp \left(-\frac{W^2}{2\sigma_0^2} \right) \\ &= \operatorname{argmax}_W W_{MLE} - \frac{W^2}{2\sigma_0^2} \\ &= \operatorname{argmax}_W W_{MLE} - \frac{\lambda}{2} W^2 \quad \lambda = \frac{1}{\sigma^2} \end{aligned}$$

We can see that under the Gaussian priori, MAP is equivalent to the linear regression with L2/ridge regularization.

Multiple Linear Regression: Fitting a line going through multiple points with two features

Plant	x1 (Age, month)	x2 (Height, cm)	y (Weight, gram)
A	1	10	1
B	3	14	2
C	4	18	2.5
...	...	...	...

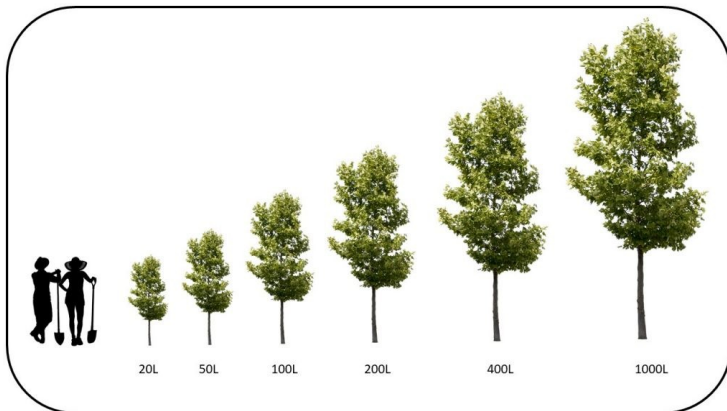
Matrix format: $y = m_1 \cdot x_1 + m_2 \cdot x_2 + c$



Multiple Linear Regression: Fitting a line going through multiple points with two features

Plant	x1 (Age, month)	x2 (Height, cm)	y (Weight, gram)
A	1	10	1
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C	4	18	2.5
...	...	...	...

Matrix format: $y = m_1x_1 + m_2x_2 + c$



$$\begin{bmatrix} 1 & 10 & 1 \\ 3 & 14 & 1 \\ 4 & 18 & 1 \\ \dots & \dots & \dots \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2.5 \\ \dots \end{bmatrix}$$

$Ax = b$

Least Squares and the Pseudoinverse

Vector $\underline{x}$ is called a *least squares solution* of linear system

$$\mathbf{A}\underline{x} = \underline{b} \quad (6)$$

when it minimizes

$$\|\mathbf{A}\underline{x} - \underline{b}\|. \quad (7)$$

For any linear system $\mathbf{A}\underline{x} = \underline{b}$, the associate normal system

$$\mathbf{A}^T \mathbf{A} \underline{x} = \mathbf{A}^T \underline{b} \quad (8)$$

is consistent, and all solutions of the normal system are least squares solutions of $\mathbf{A}\underline{x} = \underline{b}$. If $\mathbf{A}$ has linearly independent columns and if $\mathbf{A}^T \mathbf{A}$ is invertible then the system $\mathbf{A}\underline{x} = \underline{b}$ has a *unique* least squares solution:

$$\underline{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \underline{b} \quad (9)$$

If $\mathbf{A}$ does not have full rank, the least squares problem still has a solution, but it is no longer unique. There are many vectors $\underline{x}$ that minimize $\|\mathbf{A}\underline{x} - \underline{b}\|$. ***The optimal solution of the $\mathbf{A}\underline{x} = \underline{b}$ is the vector $\underline{x}$, that has the minimum length***³ This optimal solution is called $\underline{x}^+$ and the matrix which produces $\underline{x}^+$ from $\underline{b}$ is called the *pseudoinverse* of $\mathbf{A}$. Pseudoinverse of $\mathbf{A}$ is denoted by $\mathbf{A}^+$, so we have:

$$\underline{x}^+ = \mathbf{A}^+ \underline{b} \quad (10)$$

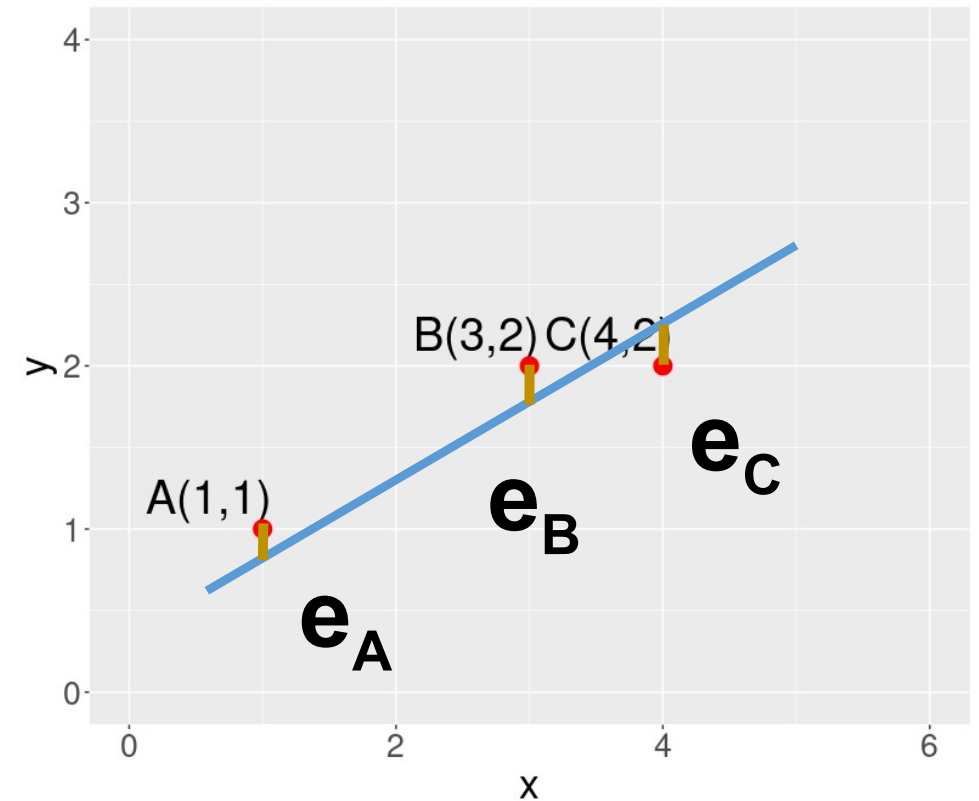
If the singular value decomposition of $\mathbf{A}$ is $\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ (eq 1), the pseudoinverse of $\mathbf{A}$ is then:

$$\mathbf{A}^+ = \mathbf{V}\mathbf{S}^+\mathbf{U}^T \quad (11)$$

The singular values $\sigma_1, \dots, \sigma_r$ are on the diagonal of m by n matrix $\mathbf{S}$, and the reciprocals of the singular values $\frac{1}{\sigma_1}, \dots, \frac{1}{\sigma_r}$ are on the diagonal of n by m matrix $\mathbf{S}^+$. *The minimum length solution to $\mathbf{A}\underline{x} = \underline{b}$ is $\underline{x}^+ = \mathbf{A}^+ \underline{b} = \mathbf{V}\mathbf{S}^+\mathbf{U}^T \underline{b}$.* The pseudoinverse of $\mathbf{A}^+$ is $\mathbf{A}^{++} = \mathbf{A}$. If $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^+ = \mathbf{A}^{-1}$ and the system⁶ has a unique least squares solution⁹

Least Squares and gradient descent

- $e_A = y_{Aline} - y_A$
- $e_B = y_{Bline} - y_B$
- $e_C = y_{Cline} - y_C$
- $e = ?$



Least Squares and gradient descent

- $e_A = y_{\text{Aline}} - y_A$
- $e_B = y_{\text{Bline}} - y_B$
- $e_C = y_{\text{Cline}} - y_C$
- $e = ?$

$$e_1^2 + e_2^2 + e_3^2 + e_4^2 = e_{\text{total}}$$
define

$$0 + 0.25 + 0.25 + 0 = 0.5$$
 ← minimum error

So the equation for error

$$e_1 = y_{1,\text{line}} - y_1$$

$$e_2 = y_{2,\text{line}} - y_2$$

$$e_3 = y_{3,\text{line}} - y_3$$

$$e_4 = y_{4,\text{line}} - y_4$$

equation for line
 $\alpha x_i + \beta = y_{\text{line}}$

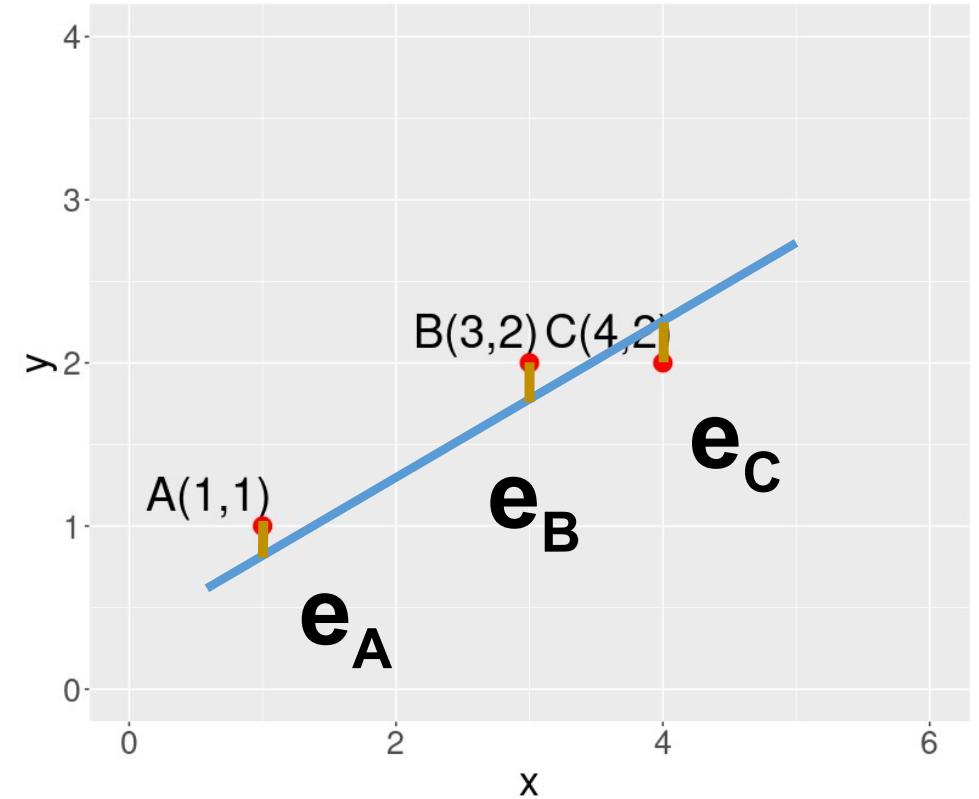
$$e_1 = [\alpha x_1 + \beta] - y_1$$

$$e_2 = [\alpha x_2 + \beta] - y_2$$

$$e_3 = [\alpha x_3 + \beta] - y_3$$

$$e_4 = [\alpha x_4 + \beta] - y_4$$

line → data.



Least Squares and gradient descent

Now we can set this problem up as an optimization gradient descent problem.

$$\min_{\alpha, \beta} \sum_{i=1}^4 (\alpha x_i + \beta - y_i)^2$$

Remember from Gradient Descent

$$\begin{bmatrix} \alpha_{\text{next}} \\ \beta_{\text{next}} \end{bmatrix} = \begin{bmatrix} \alpha_{\text{previous}} \\ \beta_{\text{previous}} \end{bmatrix} - 0.2 \begin{bmatrix} \frac{df}{d\alpha}(\alpha_{\text{previous}}) \\ \frac{df}{d\beta}(\beta_{\text{previous}}) \end{bmatrix}$$

$$f(\alpha, \beta) = \sum_{i=1}^4 (\alpha x_i + \beta - y_i)^2$$

We can pick

$$\frac{\partial f}{\partial \alpha} = \sum_{i=1}^4 2(\alpha x_i + \beta - y_i) x_i$$
$$\frac{\partial f}{\partial \beta} = \sum_{i=1}^4 2(\alpha x_i + \beta - y_i)$$
$$\begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

